

Praktikum 4 - Gruppe 10

None

None

None

Table of contents

1. Praktikum 4 - Gruppe 10	3
1.1 Data Vault to Data Mart - ETL Pipeline	3
1.2 From Data Vault to Star Schema	3
1.3 Dimensional Model Overview	3
1.4 ETL Pipeline	4
1.5 Analytical Views	5
1.6 Power BI Analytics	5
1.7 Power BI DAX Measures	8

1. Praktikum 4 - Gruppe 10

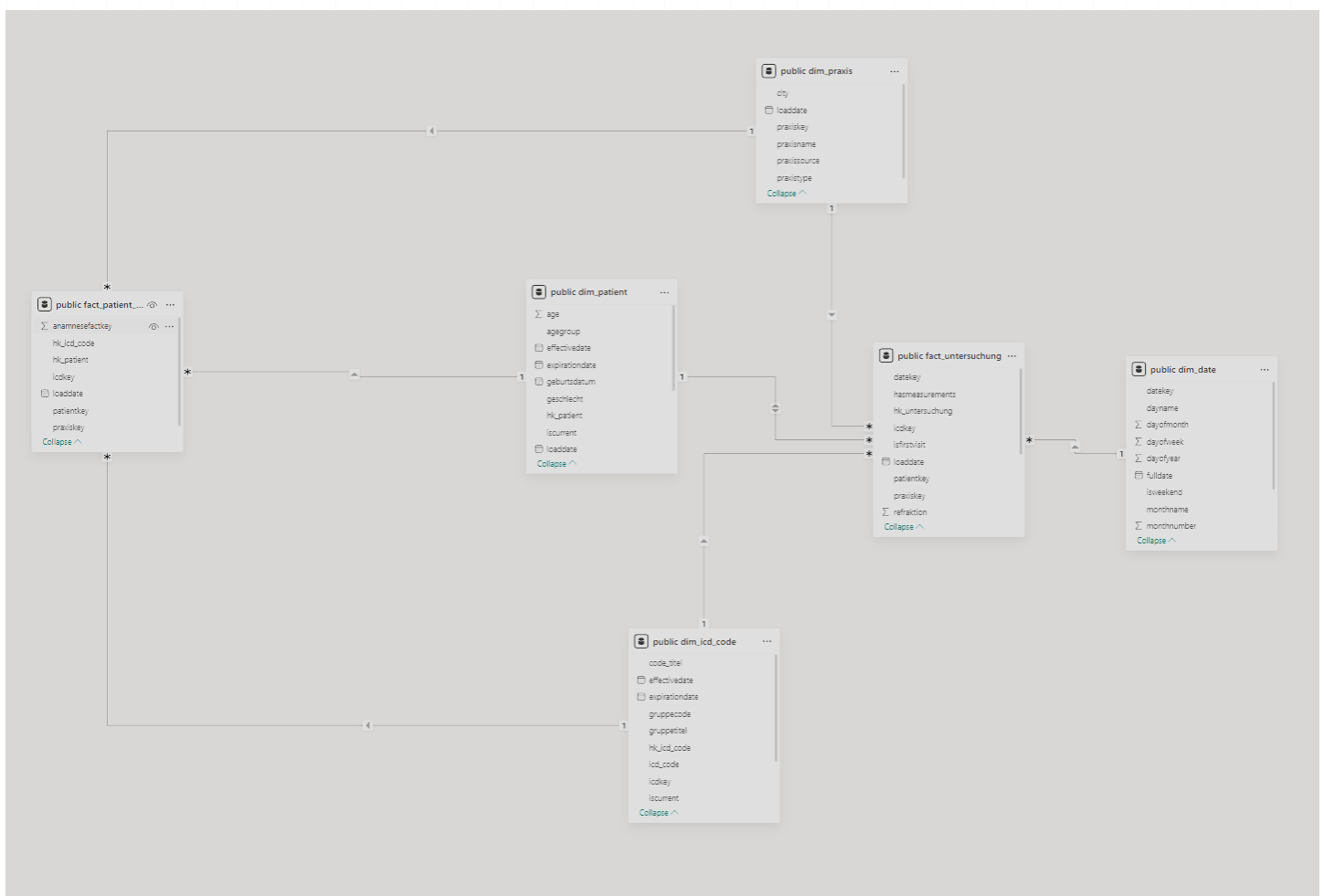
1.1 Data Vault to Data Mart - ETL Pipeline

This project transforms data from the Data Vault schema (Praktikum 3) into a dimensional model (Star Schema) optimized for analytical queries and business intelligence reporting.

1.2 From Data Vault to Star Schema

In Praktikum 3, we built a Data Vault model that preserves data lineage and handles multiple sources. Now we transform this into a dimensional model for better query performance and easier analytics.

1.3 Dimensional Model Overview



Our Star Schema consists of:

1.3.1 Dimension Tables

Dim_Date - Time dimension for temporal analysis - DateKey (PK), FullDate, DayOfWeek, DayName - Week, Month, Quarter, Year - IsWeekend flag for working day analysis

Dim_Praxis - Practice/Clinic locations - PraxisKey (PK), PraxisSource, PraxisName - PraxisType (Praxis vs Uniklinik), City - Tracks all 9 Praxis locations + 2 University Clinics

Dim_Patient - Patient master data with SCD Type 2 - PatientKey (PK), HK_Patient, PatientID, PatientSource - Demographic data: Name, Birth date, Gender, Insurance - **AgeGroup**: 5-year blocks (0-4, 5-9, 10-14, ..., 85-89, 90+) - EffectiveDate, ExpirationDate, IsCurrent for slowly changing dimensions

Dim_ICD_Code - Denormalized ICD hierarchy - ICDKey (PK), HK_ICD_Code, ICD_Code, Code_Titel - **Hierarchy**: Kapitel → Gruppe → Code (flattened) - EffectiveDate, IsCurrent for version tracking

1.3.2 Fact Tables

Fact_Untersuchung - Examination facts (grain: one examination) - Links to: Patient, ICD Code, Date, Praxis - Measurements: Tensio, Refraktion, Visus - **IsFirstVisit**: Flag for first-time patient visits (Aufgabe 2c) - HasMeasurements: Flag indicating if any measurements were taken

Fact_Patient_Anamnese - Medical history facts (grain: patient-diagnosis pair) - Links to: Patient, ICD Code, Praxis - Tracks pre-existing conditions from medical history - No date dimension (anamnese data has no timestamps)

1.4 ETL Pipeline

1.4.1 Step 1: Populate Dim_Date

Generate date records for the range 2015-2030 with all temporal attributes (day, week, month, quarter, year).

1.4.2 Step 2: Populate Dim_Praxis

Extract unique practice/clinic sources from `Hub_Patient.PatientSource` : - Parse source name to extract city and type - Distinguish between Praxis and Uniklinik - Create readable names: "Praxis Coesfeld", "Universitätsklinikum Münster"

1.4.3 Step 3: Populate Dim_ICD_Code

Denormalize the ICD hierarchy from Data Vault:

```
Hub_ICD_Kapitel → Link_Kapitel_Gruppe → Hub_ICD_Gruppe
                → Link_Gruppe_Code → Hub_ICD_Code
```

Result: Single row with Kapitel, Gruppe, and Code information (flattened)

1.4.4 Step 4: Populate Dim_Patient

Transform patient data with age calculations: - Get latest `Sat_Patient_Stammdaten` for each patient - Calculate **AgeGroup** in 5-year blocks using birth date - Calculate actual Age for detailed analysis - Mark as IsCurrent = TRUE (SCD Type 2 ready)

1.4.5 Step 5: Populate Fact_Untersuchung

Load examination facts with measurements: - Join `Hub_Untersuchung` → `Link_Patient_Untersuchung` → `Hub_Patient` - Get latest `Sat_Untersuchung` (date) and `Sat_Messwerte` (measurements) - Link to dimensions via foreign keys - Calculate **IsFirstVisit** by comparing with earliest examination date - Join to first diagnosis via

```
Link_Untersuchung_Diagnose
```

1.4.6 Step 6: Populate Fact_Patient_Anamnese

Load medical history facts: - Extract from `Link_Patient_Anamnese` - Link to Patient, ICD Code, and Praxis dimensions - Represents pre-existing conditions separate from examination diagnoses

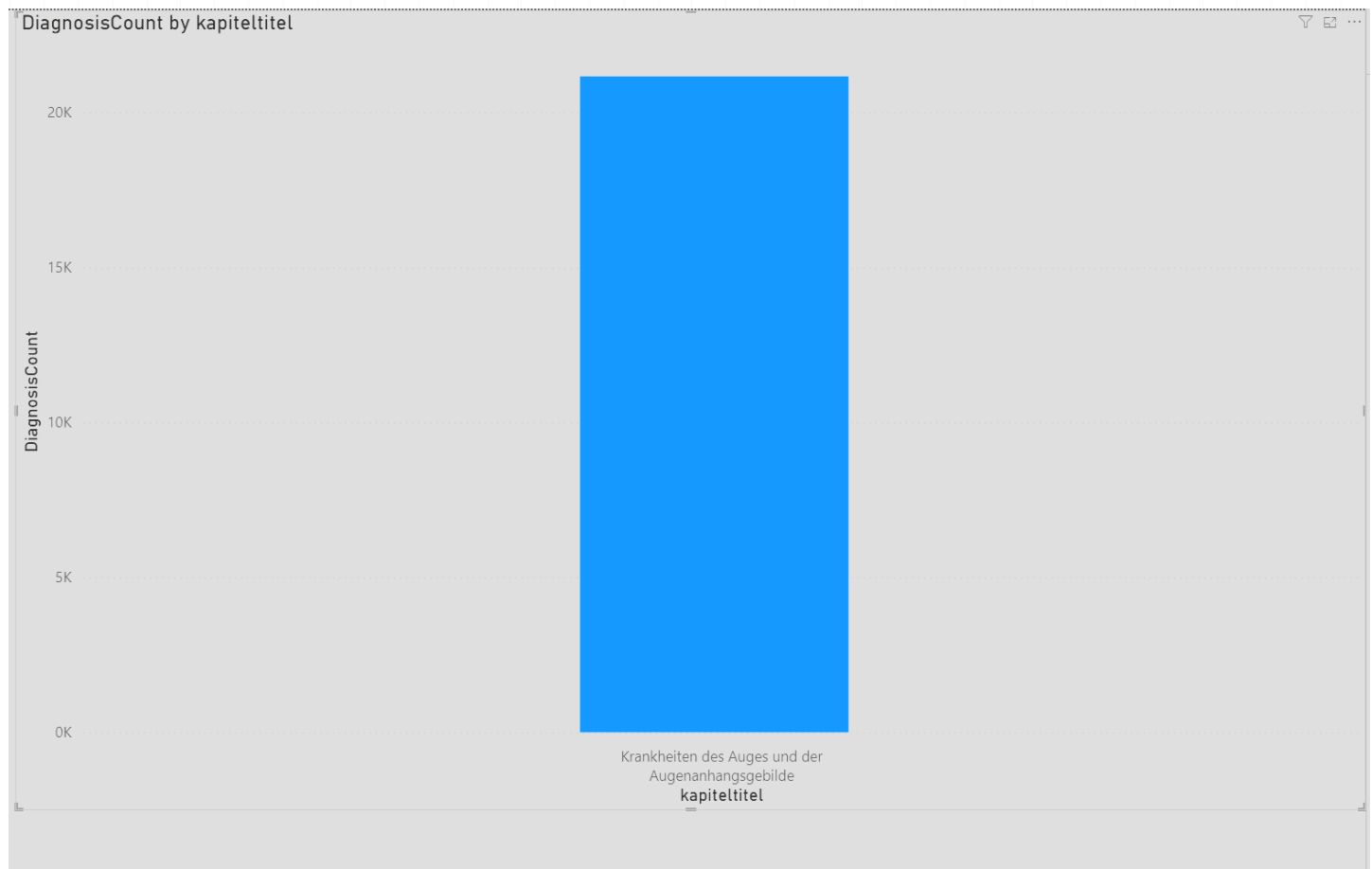
1.5 Analytical Views

Pre-built views for common analytics:

- `vw_First_Visit_Only` - First examinations only (Aufgabe 2c)
- `vw_Patient_Examination_Summary` - Complete patient examination details
- `vw_Measurements_By_AgeGroup` - Average measurements grouped by age
- `vw_Anamnese_Diagnose_Correlation` - Medical history vs diagnosis patterns
- `vw_Versicherung_Distribution` - Insurance distribution by practice type

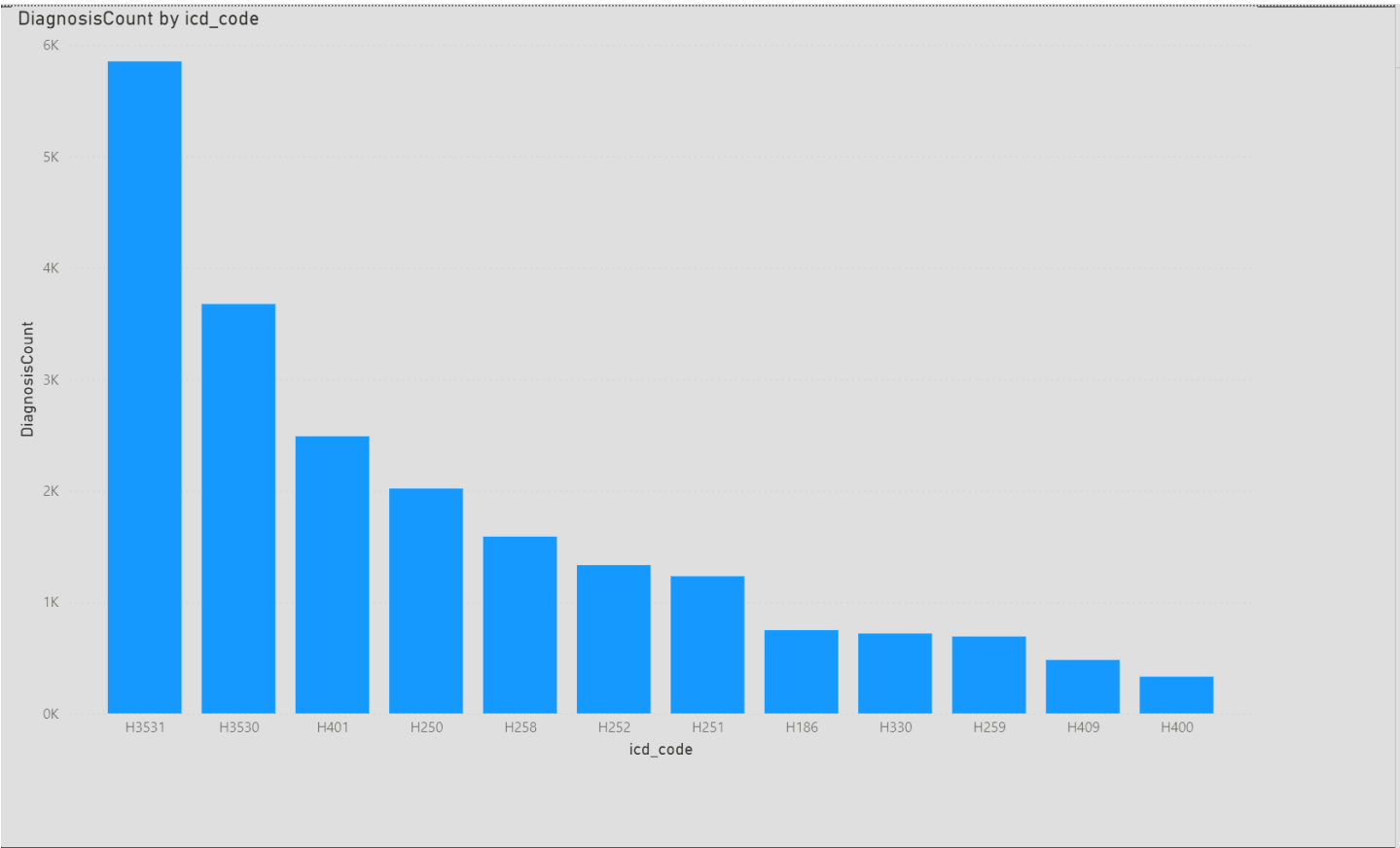
1.6 Power BI Analytics

1.6.1 Diagnosis Count by ICD Kapitel



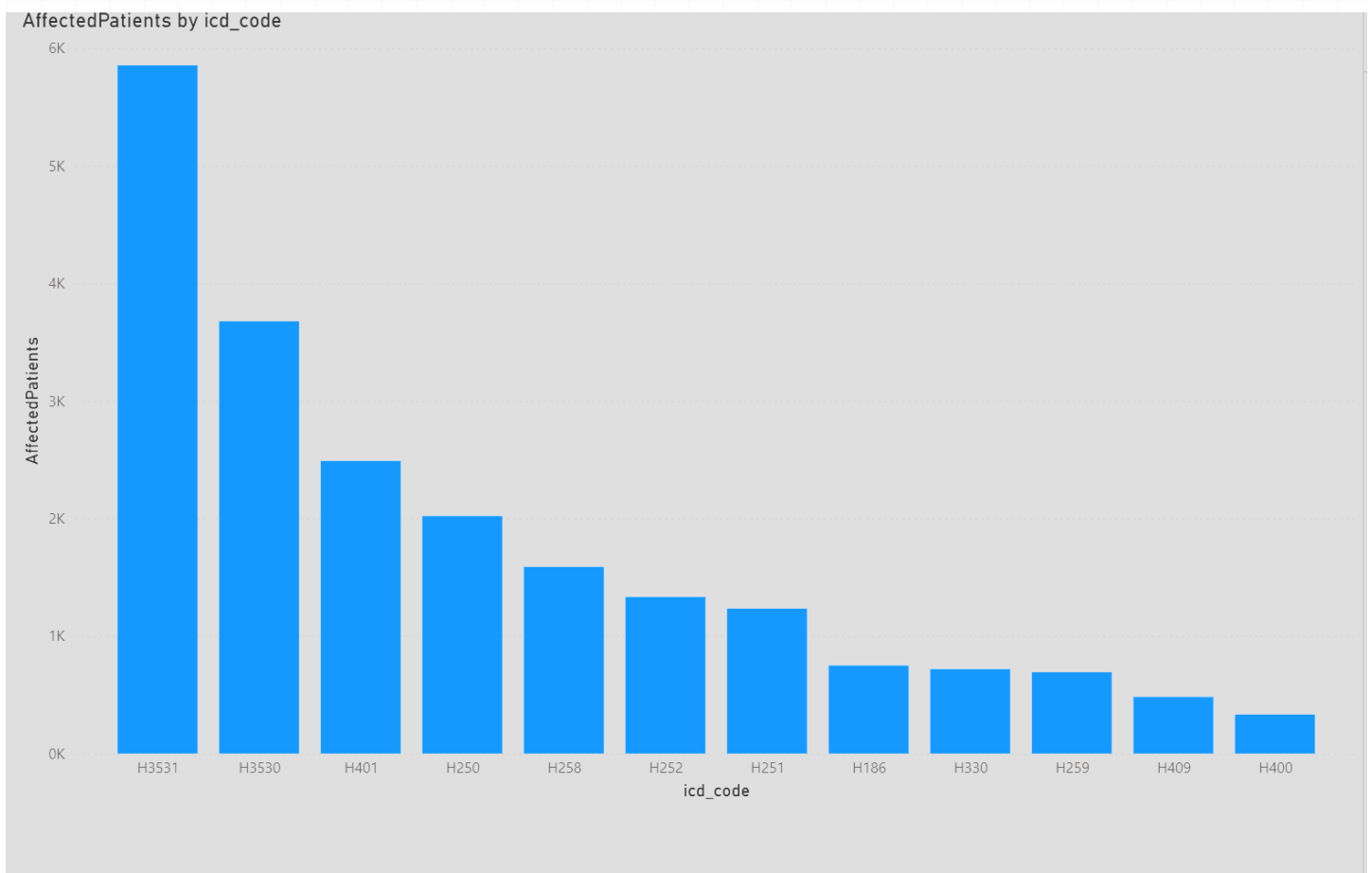
Shows the distribution of diagnoses across ICD chapters. Most common categories are eye-related conditions (Kapitel VII - Krankheiten des Auges).

1.6.2 Diagnosis Count by ICD Code



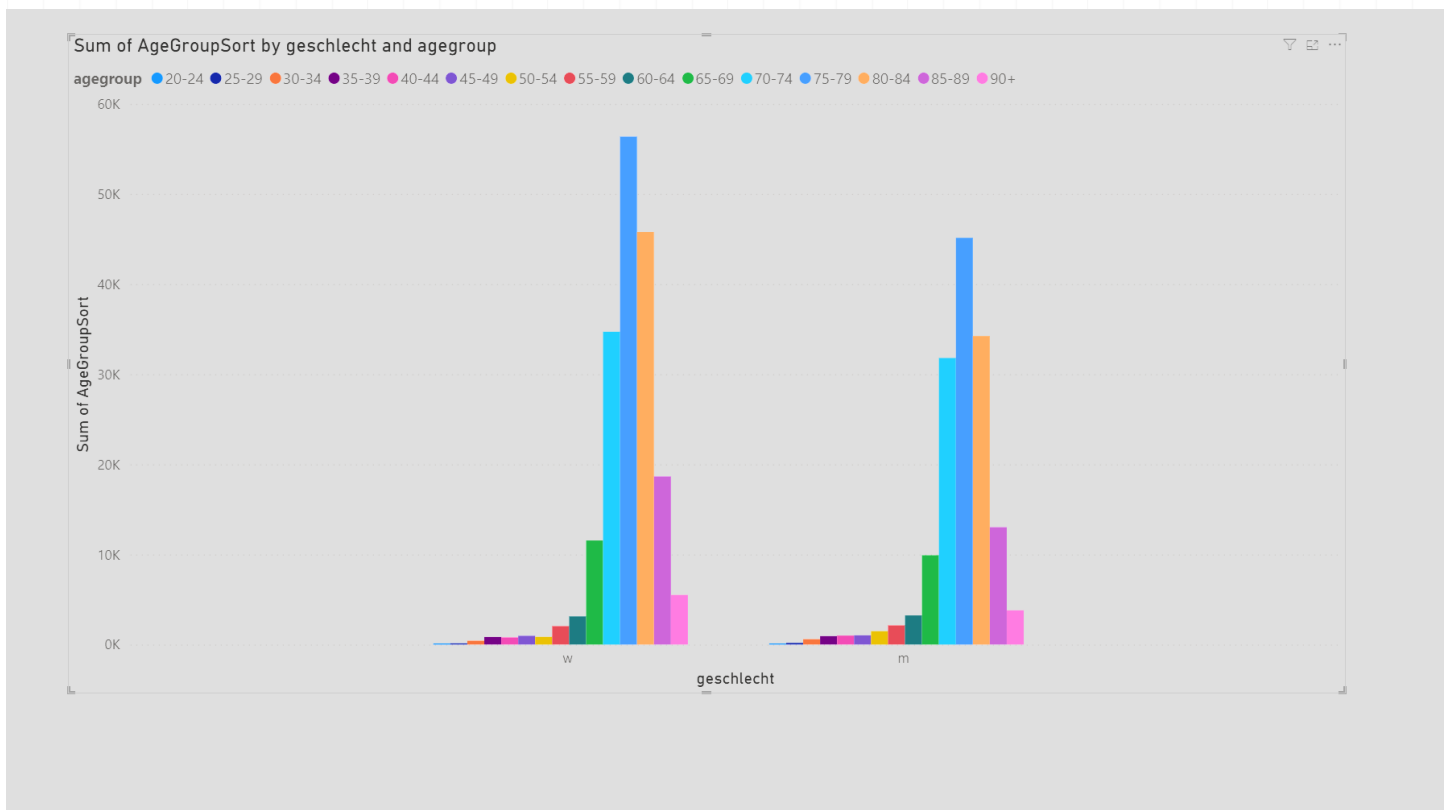
Detailed breakdown of top diagnoses at the code level, showing which specific conditions are most prevalent.

1.6.3 Affected Patients Analysis



Shows number of unique patients affected by each diagnosis, useful for understanding disease prevalence vs frequency.

1.6.4 Age Group Distribution



Patient distribution across 5-year age groups, revealing demographic patterns in the patient population.

1.7 Power BI DAX Measures

Key calculated measures for Power BI:

```
DiagnosisCount =
CALCULATE(
    COUNT('public fact_untersuchung'[UntersuchungFactKey]),
    NOT(ISBLANK('public fact_untersuchung'[ICDKey])),
    'public dim_icd_code'[IsCurrent] = TRUE
)
```

```
AffectedPatients =
CALCULATE(
    DISTINCTCOUNT('public fact_untersuchung'[PatientKey]),
    NOT(ISBLANK('public fact_untersuchung'[ICDKey])),
    'public dim_icd_code'[IsCurrent] = TRUE
)
```

These measures filter for: - Valid diagnoses (ICDKey is not blank) - Current ICD code versions only - Count examinations vs unique patients